
Calculus AB

Practice Exam

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$$(A) \quad \frac{-4x^3 + 15x^2 - 2}{(x^3 + 2)^2}$$

(B) $\frac{-2x^3 + 15x^2 + 2}{(x^3 + 2)^2}$

(C) $\frac{2x^3 - 15x^2 - 2}{(x^3 + 2)^2}$

(D) $\frac{4x^3 - 15x^2 + 2}{(x^3 + 2)^2}$

10. $\int_0^2 (x^3 + 1)^{1/2} x^2 dx =$

- (A) $\frac{52}{9}$ (B) 6 (C) $\frac{26}{3}$ (D) $\frac{52}{3}$

11. If $x^2 + xy - 3y = 3$, then at the point $(2, 1)$, $\frac{dy}{dx} =$

- (A) 5 (B) 4 (C) $\frac{7}{3}$ (D) 2

12. The number of gallons of water in a storage tank at time t , in minutes, is modeled by $w(t) = 25 - t^2$ for $0 \leq t \leq 5$. At what rate, in gallons per minute, is the amount of water in the tank changing at time $t = 3$ minutes?
- (A) 66 (B) 16 (C) -3 (D) -6

- (A) $\frac{dy}{dx} = \frac{x(4-y)}{4}$
- (B) $\frac{dy}{dx} = \frac{y(4-y)}{4}$
- (C) $\frac{dy}{dx} = \frac{xy(4-y)}{4}$
- (D) $\frac{dy}{dx} = \frac{y^2(4-y)}{4}$

14. The weight of a population of yeast is given by a differentiable function y , where $y(t)$ is measured in grams and t is measured in days. The weight of the yeast population increases according to the equation $\frac{dy}{dt} = ky$, where k is a constant. At time $t = 0$, the weight of the yeast population is 120 grams and is increasing at the rate of 24 grams per day. Which of the following is an expression for $y(t)$?
- (A) $120e^{24t}$
- (B) $120e^{t/5}$
- (C) $e^{t/5} + 119$
- (D) $24t + 120$

B**B****B****B****B****B****B****B****B**

CALCULUS AB
SECTION I, Part B

Time—45 minutes

Number of questions—15

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON THIS PART OF THE EXAM.

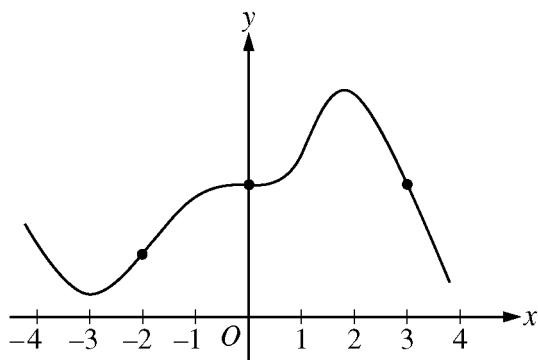
Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–90.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

B**B****B****B****B****B****B****B****B**

Graph of f

76. The graph of a differentiable function f is shown in the figure above. Which of the following is true?

- (A) $f'(-2) < f'(0) < f'(3)$
- (B) $f'(-2) < f'(3) < f'(0)$
- (C) $f'(3) < f'(-2) < f'(0)$
- (D) $f'(3) < f'(0) < f'(-2)$

B**B****B****B****B****B****B****B****B**

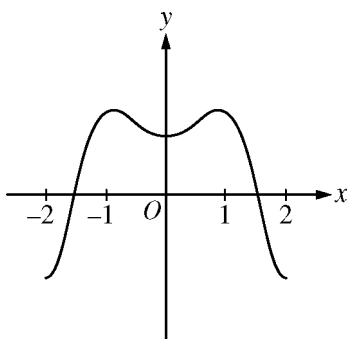
77. Let $H(x)$ be an antiderivative of $\frac{x^3 + \sin x}{x^2 + 2}$. If $H(5) = \pi$, then $H(2) =$
- (A) -9.008 (B) -5.867 (C) 4.626 (D) 12.150

-
78. The continuous function f is positive and has domain $x > 0$. If the asymptotes of the graph of f are $x = 0$ and $y = 2$, which of the following statements must be true?
- (A) $\lim_{x \rightarrow 0^+} f(x) = \infty$ and $\lim_{x \rightarrow 2} f(x) = \infty$
- (B) $\lim_{x \rightarrow 0^+} f(x) = 2$ and $\lim_{x \rightarrow \infty} f(x) = 0$
- (C) $\lim_{x \rightarrow 0^+} f(x) = \infty$ and $\lim_{x \rightarrow \infty} f(x) = 2$
- (D) $\lim_{x \rightarrow 2} f(x) = \infty$ and $\lim_{x \rightarrow \infty} f(x) = 2$

B**B****B****B****B****B****B****B****B**

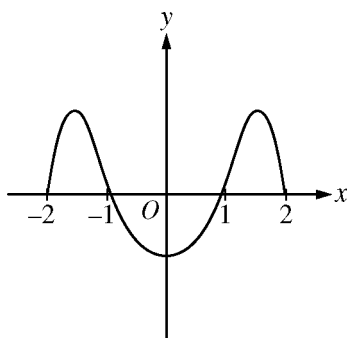
79. A file is downloaded to a computer at a rate modeled by the differentiable function $f(t)$, where t is the time in seconds since the start of the download and $f(t)$ is measured in megabits per second. Which of the following is the best interpretation of $f'(5) = 2.8$?
- (A) At time $t = 5$ seconds, the rate at which the file is downloaded to the computer is 2.8 megabits per second.
- (B) At time $t = 5$ seconds, the rate at which the file is downloaded to the computer is increasing at a rate of 2.8 megabits per second per second.
- (C) Over the time interval $0 \leq t \leq 5$ seconds, 2.8 megabits of the file are downloaded to the computer.
- (D) Over the time interval $0 \leq t \leq 5$ seconds, the average rate at which the file is downloaded to the computer is 2.8 megabits per second.
-

80. The function f has first derivative given by $f'(x) = x^4 - 6x^2 - 8x - 3$. On what intervals is the graph of f concave up?
- (A) $(2, \infty)$ only
- (B) $(0, \infty)$
- (C) $(-1, 2)$
- (D) $(-\infty, -1)$ and $(3, \infty)$

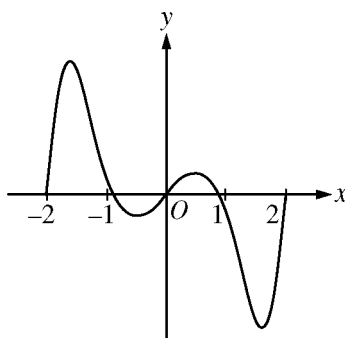
B**B****B****B****B****B****B****B****B**Graph of f

81. The graph of the function f is shown above for $-2 \leq x \leq 2$. Which of the following could be the graph of an antiderivative of f ?

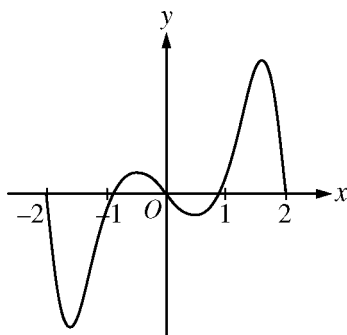
(A)



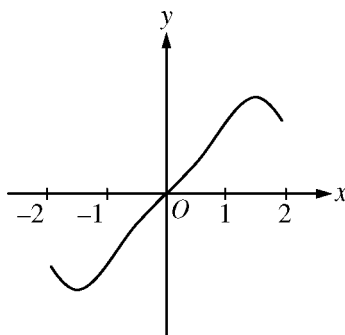
(B)



(C)



(D)



B**B****B****B****B****B****B****B****B**

82. A particle travels along a straight line with velocity $v(t) = 3e^{-t/2} \sin(2t)$ meters per second. What is the total distance, in meters, traveled by the particle during the time interval $0 \leq t \leq 2$ seconds?

- (A) 0.835 (B) 1.850 (C) 2.055 (D) 2.261

83. Let f be a function with derivative given by $f'(x) = \frac{x^3 - 8x^2 + 3}{\sqrt{x^3 + 1}}$ for $-1 < x < 9$. At what value of x does f attain a relative maximum?

- (A) -0.591 (B) 0 (C) 0.638 (D) 7.953

B**B****B****B****B****B****B****B****B**

84. The number of bacteria in a container increases at the rate of $R(t)$ bacteria per hour. If there are 1000 bacteria at time $t = 0$, which of the following expressions gives the number of bacteria in the container at time $t = 3$ hours?

(A) $R(3)$ (B) $1000 + R(3)$ (C) $\int_0^3 R(t) \, dt$ (D) $1000 + \int_0^3 R(t) \, dt$

-
85. The function g is continuous on the closed interval $[1, 4]$ with $g(1) = 5$ and $g(4) = 8$. Of the following conditions, which would guarantee that there is a number c in the open interval $(1, 4)$ where $g'(c) = 1$?

- (A) g is increasing on the closed interval $[1, 4]$.
(B) g is differentiable on the open interval $(1, 4)$.
(C) g has a maximum value on the closed interval $[1, 4]$.
(D) The graph of g has at least one horizontal tangent in the open interval $(1, 4)$.

B**B****B****B****B****B****B****B****B**

$$f''(x) = x(x - 1)^2(x + 2)^3$$

$$g''(x) = x(x - 1)^2(x + 2)^3 + 1$$

$$h''(x) = x(x - 1)^2(x + 2)^3 - 1$$

86. The twice-differentiable functions f , g , and h have second derivatives given above. Which of the functions f , g , and h have a graph with exactly two points of inflection?
- (A) g only
(B) h only
(C) f and g only
(D) f , g , and h

x	1	2	3	4	5
$f(x)$	9	4	0	-3	-5

87. The table above gives values of a function f at selected values of x . If f is twice-differentiable on the interval $1 \leq x \leq 5$, which of the following statements could be true?
- (A) f' is negative and decreasing for $1 \leq x \leq 5$.
(B) f' is negative and increasing for $1 \leq x \leq 5$.
(C) f' is positive and decreasing for $1 \leq x \leq 5$.
(D) f' is positive and increasing for $1 \leq x \leq 5$.

B**B****B****B****B****B****B****B****B**

88. Let f be the function defined by $f(x) = \ln(x^2 + 1)$, and let g be the function defined by $g(x) = x^5 + x^3$. The line tangent to the graph of f at $x = 2$ is parallel to the line tangent to the graph of g at $x = a$, where a is a positive constant. What is the value of a ?

- (A) 0.246 (B) 0.430 (C) 0.447 (D) 0.790

B**B****B****B****B****B****B****B****B**

89. Let f be a continuous function for all real numbers. Let g be the function defined by $g(x) = \int_1^x f(t) \, dt$. If the average rate of change of g on the interval $2 \leq x \leq 5$ is 6, which of the following statements must be true?

(A) The average value of f on the interval $2 \leq x \leq 5$ is 6.

(B) $g'(2) = 6$

(C) $\frac{g'(5) + g'(2)}{2} = 6$

(D) $\int_2^5 g(x) \, dx = 6$

B**B****B****B****B****B****B****B****B**

90. For any function f , which of the following statements must be true?

I. If f is defined at $x = a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.

II. If f is continuous at $x = a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.

III. If f is differentiable at $x = a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.

(A) III only

(B) I and II only

(C) II and III only

(D) I, II, and III

B**B****B****B****B****B****B****B****B**

END OF SECTION I

**IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON PART B ONLY.**

DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
- **TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET**

**AFTER TIME HAS BEEN CALLED, TURN TO PAGE 38 AND
ANSWER QUESTIONS 91–94.**

CALCULUS AB
SECTION II, Part A
Time—30 minutes
Number of questions—2

A GRAPHING CALCULATOR IS REQUIRED FOR THESE QUESTIONS.

1. A particle moves along the x -axis so that its velocity at time t is given by $v(t) = \frac{t^6 - 13t^4 + 12}{10t^3 + 3}$. At time

$t = 0$, the initial position of the particle is $x = 7$.

(a) Find the acceleration of the particle at time $t = 5.1$.

(b) Find all values of t in the interval $0 \leq t \leq 2$ for which the speed of the particle is 1.

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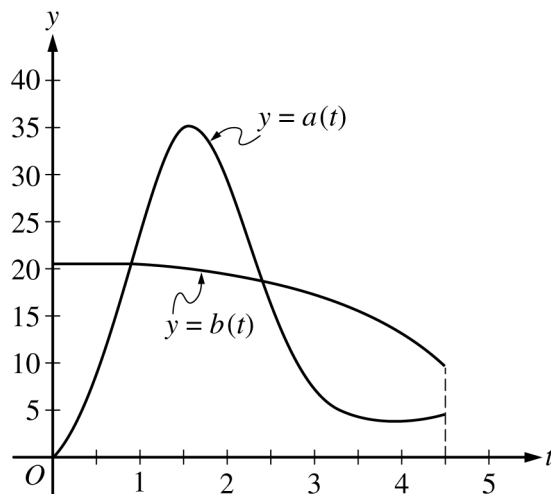
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- (c) Find the position of the particle at time $t = 4$. Is the particle moving toward the origin or away from the origin at time $t = 4$? Justify your answer.

-
- (d) During the time interval $0 < t \leq 4$, does the particle return to its initial position? Give a reason for your answer.

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2. During the time interval $0 \leq t \leq 4.5$ hours, water flows into tank A at a rate of $a(t) = (2t - 5) + 5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank B at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

(a) How much water will be in tank A at time $t = 4.5$?

- (b) During the time interval $0 \leq t \leq k$ hours, water flows into tank B at a constant rate of 20.5 liters per hour. What is the difference between the amount of water in tank A and the amount of water in tank B at time $t = k$?

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- (c) The area of the region bounded by the graphs of $y = a(t)$ and $y = b(t)$ for $k \leq t \leq 2.416$ is 14.470. How much water is in tank B at time $t = 2.416$?

-
- (d) During the time interval $2.7 \leq t \leq 4.5$ hours, the rate at which water flows into tank B is modeled by

$$w(t) = 21 - \frac{30t}{(t-8)^2} \text{ liters per hour. Is the difference } w(t) - a(t) \text{ increasing or decreasing at time}$$

$t = 3.5$? Show the work that leads to your answer.

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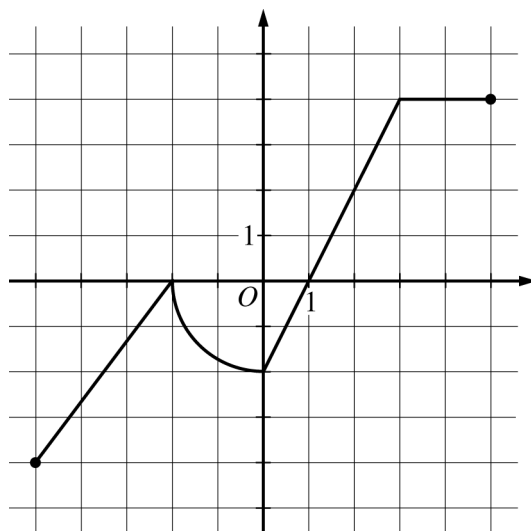
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CALCULUS AB
SECTION II, Part B
Time—1 hour
Number of questions—4

NO CALCULATOR IS ALLOWED FOR THESE QUESTIONS.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

NO CALCULATOR ALLOWED

Graph of f

3. The graph of the function f , consisting of three line segments and a quarter of a circle, is shown above. Let g be the function defined by $g(x) = \int_1^x f(t) dt$.

(a) Find the average rate of change of g from $x = -5$ to $x = 5$.

(b) Find the instantaneous rate of change of g with respect to x at $x = 3$, or state that it does not exist.

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NO CALCULATOR ALLOWED

(c) On what open intervals, if any, is the graph of g concave up? Justify your answer.

(d) Find all x -values in the interval $-5 < x < 5$ at which g has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

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NO CALCULATOR ALLOWED

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

4. The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0, 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.
- (a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

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- (b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

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NO CALCULATOR ALLOWED

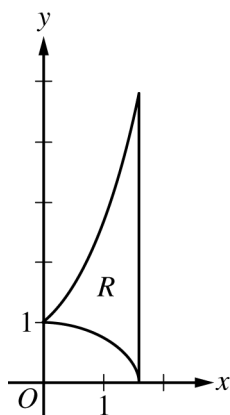
(c) Evaluate $\int_2^4 f''(x) \, dx$.

(d) Find $\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

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NO CALCULATOR ALLOWED



5. Let R be the region in the first quadrant enclosed by the graph of $f(x) = \sqrt{\cos x}$, the graph of $g(x) = e^x$, and the vertical line $x = \frac{\pi}{2}$, as shown in the figure above.

(a) Write, but do not evaluate, an integral expression that gives the area of R .

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NO CALCULATOR ALLOWED

- (b) Find the volume of the solid generated when R is revolved about the x -axis.

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- (c) Region R is the base of a solid whose cross sections perpendicular to the x -axis are semicircles with diameters on the xy -plane. Write, but do not evaluate, an integral expression that gives the volume of this solid.

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NO CALCULATOR ALLOWED

6. For $0 \leq t \leq 6$ seconds, a screen saver on a computer screen shows two circles that start as dots and expand outward.
- (a) At the instant that the first circle has a radius of 9 centimeters, the radius is increasing at a rate of $\frac{3}{2}$ centimeters per second. Find the rate at which the area of the circle is changing at that instant. Indicate units of measure.

-
- (b) The radius of the first circle is modeled by $w(t) = 12 - 12e^{-0.5t}$ for $0 \leq t \leq 6$, where $w(t)$ is measured in centimeters and t is measured in seconds. At what time t is the radius of the circle increasing at a rate of 3 centimeters per second?

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NO CALCULATOR ALLOWED

- (c) A model for the radius of the second circle is given by the function f for $0 \leq t \leq 6$, where $f(t)$ is measured in centimeters and t is measured in seconds. The rate of change of the radius of the second circle is given by $f'(t) = t^2 - 4t + 4$. Based on this model, by how many centimeters does the radius of the second circle increase from time $t = 0$ to $t = 3$?

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Answer Key for AP Calculus AB Practice Exam, Section I

Question 1: D	Question 24: C
Question 2: B	Question 25: D
Question 3: C	Question 26: C
Question 4: B	Question 27: C
Question 5: A	Question 28: B
Question 6: B	Question 29: A
Question 7: D	Question 30: B
Question 8: A	Question 76: D
Question 9: D	Question 77: B
Question 10: A	Question 78: C
Question 11: A	Question 79: B
Question 12: D	Question 80: A
Question 13: D	Question 81: D
Question 14: B	Question 82: D
Question 15: C	Question 83: C
Question 16: A	Question 84: D
Question 17: A	Question 85: B
Question 18: D	Question 86: C
Question 19: C	Question 87: B
Question 20: C	Question 88: C
Question 21: A	Question 89: A
Question 22: B	Question 90: C
Question 23: A	

2017 AP Calculus AB

Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
1	2.1C	2.1C4	Implementing algebraic/computational processes	Building notational fluency	D	91
2	3.3B(a)	3.3B3	Implementing algebraic/computational processes	Building notational fluency	B	63
3	2.1C	2.1C3	Implementing algebraic/computational processes	Building notational fluency	C	80
4	3.2B	3.2B2	Connecting multiple representations	Connecting concepts	B	68
5	2.1C	2.1C4	Implementing algebraic/computational processes	Building notational fluency	A	76
6	2.1A	2.1A1	Implementing algebraic/computational processes	Connecting concepts	B	57
7	3.2C	3.2C2	Reasoning with definitions and theorems	Building notational fluency	D	69
8	2.3B	2.3B1	Implementing algebraic/computational processes	Connecting concepts	A	68
9	3.2C	3.2C1	Connecting multiple representations	Connecting concepts	D	84
10	3.3B(b)	3.3B5	Implementing algebraic/computational processes	Building notational fluency	A	42
11	2.1C	2.1C5	Implementing algebraic/computational processes	Building notational fluency	A	61
12	2.3D	2.3D1	Connecting concepts	Implementing algebraic/computational processes	D	80
13	2.3F	2.3F1	Connecting multiple representations	Connecting concepts	D	52
14	3.5A	3.5A1	Connecting concepts	Building notational fluency	B	18
15	1.1A(b)	1.1A3	Connecting multiple representations	Connecting concepts	C	41
16	2.2A	2.2A1	Implementing algebraic/computational processes	Connecting concepts	A	60
17	2.3C	2.3C1	Implementing algebraic/computational processes	Connecting concepts	A	42
18	3.3A	3.3A2	Reasoning with definitions and theorems	Connecting concepts	D	68
19	2.3B	2.3B2	Implementing algebraic/computational processes	Connecting concepts	C	57
20	1.2A	1.2A3	Reasoning with definitions and theorems	Connecting concepts	C	24
21	2.1A	2.1A2	Connecting concepts	Building notational fluency	A	47
22	3.5A	3.5A2	Implementing algebraic/computational processes	Connecting concepts	B	39
23	3.4D	3.4D2	Connecting concepts	Building notational fluency	A	42
24	3.4B	3.4B1	Connecting concepts	Implementing algebraic/computational processes	C	44
25	2.1C	2.1C3	Connecting multiple representations	Implementing algebraic/computational processes	D	60
26	1.1C	1.1C3	Implementing algebraic/computational processes	Reasoning with definitions and theorems	C	30
27	2.2A	2.2A3	Connecting multiple representations	Connecting concepts	C	61
28	2.3C	2.3C2	Implementing algebraic/computational processes	Connecting concepts	B	46
29	1.1D	1.1D1	Building notational fluency	Connecting concepts	A	30
30	3.2A(a)	3.2A2	Implementing algebraic/computational processes	Reasoning with definitions and theorems	B	21

2017 AP Calculus AB

Question Descriptors and Performance Data

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
76	2.2A	2.2A2	Connecting multiple representations	Connecting concepts	D	81
77	3.3B(b)	3.3B2	Implementing algebraic/computational processes	Reasoning with definitions and theorems	B	55
78	1.1D	1.1D1	Building notational fluency	Connecting concepts	C	63
79	2.3A	2.3A2	Connecting concepts	Building notational fluency	B	67
80	2.2A	2.2A1	Implementing algebraic/computational processes	Connecting concepts	A	65
81	2.2A	2.2A3	Connecting multiple representations	Connecting concepts	D	49
82	3.4C	3.4C1	Implementing algebraic/computational processes	Connecting concepts	D	42
83	2.2A	2.2A1	Implementing algebraic/computational processes	Connecting concepts	C	72
84	3.4E	3.4E1	Connecting concepts	Building notational fluency	D	89
85	2.4A	2.4A1	Reasoning with definitions and theorems	Connecting concepts	B	63
86	2.2A	2.2A1	Connecting concepts	Connecting multiple representations	C	44
87	2.2A	2.2A2	Connecting multiple representations	Connecting concepts	B	44
88	2.3B	2.3B1	Implementing algebraic/computational processes	Connecting concepts	C	61
89	3.4B	3.4B1	Reasoning with definitions and theorems	Connecting concepts	A	37
90	1.2A	1.2A1	Reasoning with definitions and theorems	Connecting concepts	C	54

Free-Response Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus	Mean
1	1.2B 2.3C 3.4C	1.2B1 2.3C1 3.4C1	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Building notational fluency Communicating	3.04
2	2.3D 3.2C 3.4A 3.4D	2.3D1 3.2C2 3.4A2 3.4D1	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Connecting multiple representations Building notational fluency Communicating	3.01
3	2.1A 2.2A 3.2C 3.3A	2.1A1 2.2A1 3.2C1 3.3A2,3.3A3	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Connecting multiple representations Building notational fluency Communicating	3.49
4	1.1C 2.1C 3.2B 3.3B(b)	1.1C3 2.1C2 3.2B2 3.3B2	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Connecting multiple representations Building notational fluency Communicating	2.44
5	3.3B(b) 3.4D	3.3B2,3.3B5 3.4D1,3.4D2	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Connecting multiple representations Building notational fluency Communicating	4.46
6	2.1C 2.3A 2.3C 3.1A 3.3B(b) 3.4A	2.1C2,2.1C4 2.3A1 2.3C2 3.1A2 3.3B2 3.4A2	Reasoning with definitions and theorems Connecting concepts Implementing algebraic/computational processes Building notational fluency Communicating	5.39

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Question 1

(a) $v'(5.1) = 6.491816$

The acceleration of the particle at time $t = 5.1$ is 6.492 (or 6.491).

1 : answer

(b) $|v(t)| = 1$ for $0 \leq t \leq 2 \Rightarrow t = 0.771829, t = 1.400556$

The speed of the particle is 1 at times $t = 0.772$ (or 0.771) and $t = 1.401$ (or 1.400).

2 : $\begin{cases} 1 : \text{considers } |v(t)| = 1 \\ 1 : \text{answer} \end{cases}$

(c) Let $x(t)$ be the position of the particle at time t .

$$x(4) = 7 + \int_0^4 v(t) dt = 6.711558$$

The position of the particle at time $t = 4$ is 6.712 (or 6.711).

The particle is moving away from the origin because $x(4) > 0$ and $v(4) = 1.213064 > 0$.

4 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{uses initial condition} \\ 1 : \text{position} \\ 1 : \text{movement relative to the origin, with justification} \end{cases}$

(d) $x'(t) = v(t) \Rightarrow x(t)$ is continuous on $0 < t \leq 4$.

$$x(1) = 7 + \int_0^1 v(t) dt = 9.403593$$

From part (c), $x(4) = 6.712$.

From time $t = 1$ to $t = 4$, the particle moves from $x = 9.404$ to $x = 6.712$. By the Intermediate Value Theorem, the particle must return to $x = 7$ during the time interval.

2 : $\begin{cases} 1 : \text{sandwiches initial position} \\ 1 : \text{answer with reason} \end{cases}$

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Question 2

(a) $\int_0^{4.5} a(t) \, dt = 66.532128$

At time $t = 4.5$, tank A contains 66.532 liters of water.

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(b) $a(k) = 20.5 \Rightarrow k = 0.892040$

$$\int_0^k (20.5 - a(t)) \, dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

3 : $\begin{cases} 1 : \text{sets } a(k) = 20.5 \\ 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(c) $\int_0^{2.416} b(t) \, dt = \int_0^k b(t) \, dt + \int_k^{2.416} b(t) \, dt$

$$\int_0^k b(t) \, dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t) \, dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B .

Therefore, $\int_k^{2.416} b(t) \, dt = \int_k^{2.416} a(t) \, dt - 14.470 = 30.027051$.

$$\int_0^k b(t) \, dt + \int_k^{2.416} b(t) \, dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

2 : $\begin{cases} 1 : \int_k^{2.416} a(t) \, dt \\ 1 : \text{answer} \end{cases}$

(d) $w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

2 : $\begin{cases} 1 : w'(3.5) - a'(3.5) < 0 \\ 1 : \text{conclusion} \end{cases}$

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Question 3

(a) $\frac{g(5) - g(-5)}{5 - (-5)} = \frac{12 - (\pi + 7)}{10} = \frac{5 - \pi}{10}$

3 : $\begin{cases} 1 : \text{difference quotient} \\ 2 : \text{answer} \end{cases}$

(b) $g'(x) = f(x)$
 $g'(3) = f(3) = 4$

1 : answer

The instantaneous rate of change of g at $x = 3$ is 4.

(c) The graph of g is concave up on $-5 < x < -2$ and $0 < x < 3$,
because $g'(x) = f(x)$ is increasing on these intervals.

2 : intervals with justification

(d) $g'(x) = f(x)$ is defined at all x with $-5 < x < 5$.

$g'(x) = f(x) = 0$ at $x = -2$ and $x = 1$.

Therefore, g has critical points at $x = -2$ and $x = 1$.

g has neither a local maximum nor a local minimum at $x = -2$
because g' does not change sign there.

g has a local minimum at $x = 1$ because g' changes from negative to
positive there.

3 : $\begin{cases} 1 : \text{considers } f(x) = 0 \\ 1 : \text{critical points at} \\ \quad x = -2 \text{ and } x = 1 \\ 1 : \text{answers with justifications} \end{cases}$

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Question 4

(a) $\int_0^6 f'(x) \, dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$

$$f(6) - f(0) = \int_0^6 f'(x) \, dx$$

$$f(6) = f(0) + \int_0^6 f'(x) \, dx \approx 20 + 20.2 = 40.2$$

3 : $\begin{cases} 1 : \text{midpoint sum} \\ 1 : \text{Fundamental Theorem} \\ \quad \text{of Calculus} \\ 1 : \text{answer} \end{cases}$

(b) Since $f'(x) \leq 7$, $\int_0^6 f'(x) \, dx \leq 6 \cdot 7 = 42$.

$$f(6) - f(0) \leq 42 \Rightarrow f(6) \leq 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

2 : $\begin{cases} 1 : \text{integral bound} \\ 1 : \text{answer with reasoning} \end{cases}$

(c) $\int_2^4 f''(x) \, dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$

2 : $\begin{cases} 1 : \text{Fundamental Theorem} \\ \quad \text{of Calculus} \\ 1 : \text{answer} \end{cases}$

(d) $\lim_{x \rightarrow 0} (f(x) - 20e^x) = 0$

$$\lim_{x \rightarrow 0} (0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 0} \frac{f(x) - 20e^x}{0.5f(x) - 10} = \lim_{x \rightarrow 0} \frac{f'(x) - 20e^x}{0.5f'(x)} = \frac{4 - 20}{0.5(4)} = -8$$

2 : $\begin{cases} 1 : \text{L'Hospital's Rule} \\ 1 : \text{answer} \end{cases}$

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Question 5

(a) $\text{Area} = \int_0^{\pi/2} (g(x) - f(x)) \, dx$

2 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{limits} \end{cases}$

(b) $\text{Volume} = \pi \int_0^{\pi/2} ((g(x))^2 - (f(x))^2) \, dx$
 $= \pi \int_0^{\pi/2} ((e^x)^2 - (\sqrt{\cos x})^2) \, dx$
 $= \pi \int_0^{\pi/2} (e^{2x} - \cos x) \, dx = \pi \left[\frac{1}{2} e^{2x} - \sin x \right]_{x=0}^{x=\pi/2}$
 $= \pi \left(\frac{1}{2} e^{\pi} - \sin \frac{\pi}{2} - \left(\frac{1}{2} - 0 \right) \right) = \pi \left(\frac{1}{2} e^{\pi} - \frac{3}{2} \right)$

4 : $\begin{cases} 1 : \text{integrand} \\ 2 : \text{antiderivatives} \\ 1 : \text{answer} \end{cases}$

(c) $\text{Volume} = \frac{1}{2} \int_0^{\pi/2} \pi \left(\frac{g(x) - f(x)}{2} \right)^2 \, dx$

3 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{limits} \\ 1 : \text{constant} \end{cases}$

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Question 6

(a) $A = \pi r^2$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\left. \frac{dA}{dt} \right|_{r=9} = 2\pi \cdot 9 \cdot \frac{3}{2} = 27\pi$$

When the radius is 9 centimeters, the area is changing at a rate of $27\pi \text{ cm}^2/\text{sec}$.

(b) $w'(t) = (-12)(-0.5)e^{-0.5t} = 6e^{-0.5t}$

$$6e^{-0.5t} = 3 \Rightarrow e^{-0.5t} = \frac{1}{2} \Rightarrow -0.5t = \ln\left(\frac{1}{2}\right) \Rightarrow t = 2\ln 2$$

The radius is increasing at a rate of 3 centimeters per second at time $t = 2\ln 2$ seconds.

(c) $\int_0^3 (t^2 - 4t + 4) dt = \left[\frac{1}{3}t^3 - 2t^2 + 4t \right]_{t=0}^{t=3}$

$$= \left(\frac{1}{3} \cdot 3^3 - 2 \cdot 3^2 + 4 \cdot 3 \right) - \left(\frac{1}{3} \cdot 0^3 - 2 \cdot 0^2 + 4 \cdot 0 \right)$$

$$= 9 - 18 + 12 = 3$$

The radius increases by 3 centimeters from time $t = 0$ to time $t = 3$ seconds.

$$3 : \begin{cases} 1 : \frac{dA}{dt} \\ 1 : \text{answer} \\ 1 : \text{units} \end{cases}$$

$$3 : \begin{cases} 1 : w'(t) \\ 1 : \text{sets } w'(t) = 3 \\ 1 : \text{answer} \end{cases}$$

$$3 : \begin{cases} 1 : \text{integral} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$$